

Wireshark

Category: Network Traffic Analysis • Packet Capture

Purpose: Learn how Wireshark captures, filters, and analyzes network traffic and how the results can support troubleshooting, security monitoring, and incident investigation.

1. What is Wireshark?

Wireshark is a network protocol analyzer that captures network packets and presents their contents in a human-readable form. It is commonly used to troubleshoot networks, understand protocols, analyze packet captures, and investigate suspicious traffic.

2. Where Wireshark Fits in Cybersecurity

Area	Use
Traffic analysis	Inspect packets, protocols, endpoints, and communication patterns.
Incident response	Investigate packet captures for suspicious or unusual activity.
Troubleshooting	Find connection failures, retransmissions, DNS problems, and protocol errors.
Learning	Understand how protocols such as TCP, DNS, HTTP, and TLS appear on the wire.

3. Basic Workflow

- Choose the correct network interface or open an existing capture file.
- Start a capture when you have authorization and a clear analysis objective.
- Generate or observe the traffic you want to study.
- Stop the capture and apply display filters.
- Inspect packets, protocol fields, conversations, and endpoints.
- Correlate packet-level findings with logs and other security evidence.

4. Capture vs Display Filters

Filter type	Purpose	Example
Capture filter	Limits what is captured.	host 192.168.13.10
Display filter	Filters packets already captured.	ip.addr == 192.168.13.10

5. Useful Display Filters

`ip.addr == 192.168.13.10`

Show packets involving a specific IP address.

`tcp`

Show TCP packets.

`udp`

Show UDP packets.

`dns`

Show DNS traffic.

`http`

Show HTTP traffic when Wireshark recognizes the protocol as HTTP.

`tcp.port == 443`

Show TCP traffic using port 443.

`tcp.flags.syn == 1`

Find TCP packets with the SYN flag set.

6. Packet Details

Area	What to examine
Frame	Capture metadata such as frame length and arrival information.
Ethernet	Source and destination MAC addresses.
IP	Source/destination IP addresses and packet-level fields.
TCP/UDP	Ports, flags, sequence information, and transport behavior.
Application protocol	Protocol-specific fields such as DNS queries or HTTP information.

7. TCP Analysis

- SYN, SYN-ACK, and ACK can reveal TCP connection establishment.
- RST may indicate connection rejection or termination.
- Retransmissions can indicate packet loss or network problems.
- Sequence and acknowledgement information helps understand TCP flow.
- TCP streams can reconstruct an application conversation when the traffic is available and not protected by encryption.

8. Common Security Analysis Tasks

- Identify unexpected connections between hosts.
- Investigate unusual DNS queries or repeated connection attempts.
- Look for suspicious protocol usage or unexpected destinations.
- Examine cleartext protocols carefully because sensitive information may be visible.
- Use packet timestamps and conversations to build an incident timeline.

9. Following a TCP Stream

Wireshark can reconstruct the application-level conversation carried by a TCP stream when the relevant packets are present. This is useful for understanding a session, but encryption such as TLS can prevent the application payload from being directly readable.

10. Common Mistakes

- Using a display filter when a capture filter was intended, or vice versa.
- Assuming every packet contains an application payload.
- Treating an IP address or port as proof of an application identity.
- Ignoring time, direction, and surrounding packets.
- Expecting encrypted TLS traffic to reveal plaintext application data without the required decryption context.

11. Practical Lab

In your VirtualBox lab, capture traffic between your Kali and intentionally vulnerable test VMs. Start with simple traffic such as ICMP, DNS, and TCP connections. Apply filters, inspect the protocol layers, and document what changes when a connection is established, used, and closed.

12. Quick Revision

Concept	Remember
Wireshark	Packet and network protocol analyzer.
Capture filter	Controls what packets are captured.
Display filter	Controls which captured packets are shown.
TCP stream	A sequence of packets belonging to a TCP conversation.
Protocol analysis	Examines packet fields and communication behavior.

13. Key Takeaways

- Wireshark turns packet captures into analyzable protocol information.

- Filters are essential for reducing large captures to relevant traffic.
- Packet analysis is useful for both troubleshooting and security investigations.
- Always capture and inspect traffic only where you are authorized to do so.